Proposal of

STYLE-AWARE NEURAL MACHINE TRANSLATION OF LOW-RESOURCE TEXTS USING LARGE LANGUAGE MODELS AND REINFORCEMENT LEARNING

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ABSTRACT

Neural machine translation of literary and scriptural texts remains an open challenge, particularly for low-resource language pairs where preserving an authoritative translator's stylistic register is as important as semantic accuracy. Current large language model (LLM) based systems produce fluent output but tend to default to a neutral, generic register, which is inadequate for literary and sacred corpora where formality, cadence, and lexical choice carry meaning. Despite recent progress in style-aware translation, prompting, and reinforcement learning, to our knowledge no published study systematically compares these adaptation strategies on the combined challenge of a low-resource, mixed Persian and Arabic literary corpus translated into a stylistically constrained English register.

This project addresses that gap through a comparative experimental study of three LLM adaptation strategies applied to a parallel corpus of Bahá'í scriptures. Parameter-Efficient Fine-Tuning (PEFT/LoRA) to establish a domain-adapted baseline, Adaptive Few-Shot Prompting (AFSP) to examine inference-time stylistic control, and a bounded Reinforcement Learning stage using a multi-component reward function combining semantic, lexical, and style-sensitive signals. The unadapted base model is included as a reference point. Translation quality is assessed through a combined framework of neural semantic metrics, objective stylometric features, and LLM-as-Judge scoring on a held-out test set. The expected contribution is an empirical comparison of these strategies for preserving Shoghi Effendi's formal scriptural register under the resource constraints of an undergraduate project.

1 INTRODUCTION

1.1 SUMMARY OF THE FIELD OF THE STUDY AND SPECIFIC RESEARCH PROBLEM, THE SIGNIFICANCES OF THE RESEARCH PROBLEM AND THE MOTIVATION

Machine Translation (MT) has progressed from rule-based and statistical systems to neural machine translation (NMT) and, more recently, to translation pipelines built on Large Language Models (LLMs) based on the Transformer architecture [6, 7]. Contemporary systems produce fluent and contextually coherent output and are now widely used in research and industry. However, the dominant design and evaluation paradigms target general domains such as news, web text, and everyday communication, where sentence-level adequacy and fluency are the primary objectives. As a result, these systems address stylistic and literary properties only indirectly, even though such properties are central to a substantial class of real-world translation tasks [1, 4, 5].

This limitation is particularly consequential for literary and religious texts, including the mixed Persian and Arabic writings of the Bahá'í corpus. In such works, vocabulary, rhythm, tone, and level of formality are not decorative features but constitutive elements of meaning [4, 5]. The authorized English translations by Shoghi Effendi are recognized for a dignified, formal, and at times archaic scriptural register. General-purpose MT systems typically yield output that is grammatically acceptable but stylistically neutral, with the result that the distinctive translator-specific voice — and the literary effect associated with it — is not reliably preserved [1, 7].

Preliminary observations on the corpus used in this project, reported in Section 9, are consistent with this pattern across several current commercial LLMs.

The specific research problem addressed in this project is therefore how LLM adaptation techniques (namely Parameter-Efficient Fine-Tuning (PEFT), Adaptive Few-Shot Prompting (AFSP), and a bounded Reinforcement Learning (RL) stage) can be combined and compared for the task of translating literary Persian and mixed Persian–Arabic texts into English while preserving both semantic content and a target translator-specific register.

The significance of this problem is twofold. From a practical perspective, many readers encounter Persian and Arabic literary and sacred texts primarily through English translation; stylistic loss in translation therefore directly affects how these texts are received. From a scientific perspective, the problem brings together several open questions in NLP, including low-resource adaptation, stylistic control, and the evaluation of literary translation quality areas in which recent shared-task findings indicate that MT remains unresolved [10]. The motivation of this project is to examine, through a comparative study, whether and to what extent these adaptation techniques can be combined to support style-aware literary translation in a low-resource setting, and to document the trade-offs that arise across the three strategies.

1.2 Summary of the literature review, the research gap and the research aim

The preliminary literature review confirms substantial progress in NMT and LLM-based translation, including work on style-aware translation and text style transfer [1, 5]. Existing studies indicate that neural systems can, in principle, model stylistic patterns and respond to style-related signals or in-context examples [5, 6], and that LLMs can outperform dedicated MT systems in creative or literary domains when guided by carefully designed prompts and reference exemplars [7, 19]. At the same time, recent shared-task findings report that MT performance remains uneven in low-resource and stylistically demanding settings [10].

Three interconnected gaps emerge from this body of work and motivate the present project.

First, the literature does not contain a systematic comparison of the three main LLM adaptation strategies applied to the same low-resource literary translation task. Individual studies typically examine one strategy in isolation [1, 8], which limits what can be concluded about the relative contribution of each method to stylistic fidelity under matched conditions.

Second, work on LLM adaptation for low-resource literary translation involving Persian and Arabic source material is limited. Most published evaluations concentrate on high-resource language pairs or on general domains. Mixed-language source passages, in which Persian and Arabic occur within the same document and must be rendered into a stylistically coherent English register, are rarely treated explicitly [9].

Third, reference-based metrics such as BLEU and TER are known to be insufficient on their own for assessing tone, register, and literary appropriateness [11]. More recent style-sensitive evaluation approaches including LLM-as-Judge protocols [12, 15] and objective stylometric measures such as lexical density and type–token ratio have been proposed, but their behavior on low-resource literary translation is not yet well characterized [14, 15]. The present project applies these methods together to evaluate the three adaptation strategies.

In view of these gaps, the aim of this project is to conduct a comparative experimental study of LLM adaptation strategies for style-aware translation of low-resource Persian and Arabic literary texts into English. Specifically, the project will construct a domain-adapted baseline using Parameter-Efficient Fine-Tuning (PEFT/LoRA); develop a retrieval-based Adaptive Few-Shot Prompting pipeline for inference-time stylistic control; conduct a bounded Reinforcement Learning experiment using a multi-component reward combining semantic, lexical, and style-sensitive signals; and evaluate all systems within a combined framework of neural semantic metrics, objective stylometric features, and LLM-as-Judge scoring. The intended contribution is an empirical comparison of these strategies for preserving Shoghi Effendi's formal scriptural register under the resource constraints of an undergraduate project.

1.3 Summary of the proposed methodology and the expected results

The methodology, described in detail in Section 6, is structured as a three-way comparison of LLM adaptation strategies applied to the same open-source base model and the same parallel corpus of Bahá'í scriptures a domain-adapted baseline using Parameter-Efficient Fine-Tuning (PEFT/LoRA), a retrieval-based Adaptive Few-Shot Prompting (AFSP) pipeline, and a bounded Reinforcement Learning stage with a multi-component style-sensitive reward. The unadapted base model is included as a reference point. All systems are evaluated within a combined framework of neural semantic metrics, objective stylometric features, and LLM-as-Judge scoring on a held-out test set.

The expected outcome, discussed in Section 7, is an empirical comparison of the three strategies in terms of semantic accuracy, stylistic fidelity, and compute cost on a low-resource literary translation task. The comparison is intended to identify which strategy, under the resource constraints of an undergraduate project, offers the most workable trade-off for preserving Shoghi Effendi's formal scriptural register.

2 Problem Statement

The research problem addressed in this project is how Large Language Model (LLM) adaptation techniques can be combined and compared for style-aware translation of low-resource literary texts, with scriptures in mixed Persian and Arabic as the source domain and the formal scriptural register of Shoghi Effendi's authorized English translations as the target style. As outlined in Section 1.1, standard MT systems produce fluent translations but tend toward a stylistically neutral register, which is inadequate for texts in which stylistic features are constitutive of meaning [1, 5].

For an LLM-based system to address this task, three computational challenges must be handled together.

Stylistic conditioning: the translation function must depend not only on the source text but on a target stylistic register, and the relevant stylistic features (formality, archaism, cadence, lexical choice) are largely non-lexical and difficult to capture with token-level supervised objectives [1, 5].

Low-resource adaptation: the available parallel corpus is small, and the source documents contain mixed-language passages that are uncommon in standard multilingual training data [9]. The third is

Evaluation under metric inadequacy: reference-based metrics such as BLEU correlate poorly with stylistic quality in literary translation, and replacement metrics (including LLM-as-Judge scoring and stylometric analysis) have known limitations of their own in this setting [11, 15].

These three challenges interact. A method that handles one well may handle the others poorly, for example, supervised fine-tuning addresses low-resource adaptation but optimizes a token-level loss that does not directly reward stylistic fidelity; in-context prompting avoids parameter updates but is sensitive to the quality of available exemplars [8]; and reinforcement learning can target non-differentiable stylistic objectives but introduces training stability and compute concerns. The research problem is therefore not the choice of a single best method but the empirical characterization of how three established adaptation strategies trade off these three challenges on the same task and the same corpus.

The remainder of this section formalizes the problem (Section 2.1), discusses its significance (Section 2.2), identifies the specific gaps the project addresses (Section 2.3), and states the research questions, objectives, and hypothesis that follow (Section 2.4).

2.1 Mathematical formulation of the problem / formal definition

The task is to learn a style-aware translation function that, given a literary source segment and a desired translator style, produces an English translation that is both semantically faithful to the source and stylistically aligned with the target register.

We define the following:

- X: the set of possible source segments in Persian and mixed Persian–Arabic (sentences or short passages).
- Y: the set of possible target segments in English.
- $x \in X$: a given source segment.
- $y \in Y$: a candidate English translation of x.
- $y^* \in Y$: the reference (human-authored) English translation of x.
- S: the set of stylistic registers under consideration (e.g., the formal scriptural register of Shoghi Effendi, a neutral register).
- $S_T \in S$: the target style for a given experiment. In this project, S_T is fixed as Shoghi Effendi's authorized register.
- P_D: a parallel corpus for the domain D, consisting of pairs (x, y^*).
- M_θ: the translation model (an open-source LLM) with parameters θ.

The model is required to learn a conditional mapping

$$M_\theta: (x, S_T) \mapsto y, \text{equivalently} P_\theta(y \mid x, S_T).$$

Quality components

Translation quality is assumed to decompose into three components:

1. **Fidelity** — the degree to which $y$preserves the meaning of x.
2. **StyleMatch** — the degree to which $y$matches the target register S_T.
3. **Fluency** — the grammaticality and naturalness of $y$in English.

A utility function Ucombines these into a scalar:

$$U(y \mid x, S_T) = \alpha_1 \cdot \text{Fidelity}(y, x) + \alpha_2 \cdot \text{StyleMatch}(y, S_T) + \alpha_3 \cdot \text{Fluency}(y),$$

with non-negative weights $\alpha_1, \alpha_2, \alpha_3 \geq 0$. The training objective is to choose parameters that maximize expected utility on the corpus:

$$\theta^* = \arg\max_\theta \ \mathbb{E}_{(x,y^*)\sim P_D}[U(y \mid x, S_T)].$$

Reinforcement learning formulation

For the bounded Reinforcement Learning stage described in Section 6, the utility components are operationalized as a scalar reward $r(y)$combining semantic, lexical, and style-sensitive signals:

$$r(y) = \omega_1 \cdot \text{COMET}(x, y) + \omega_2 \cdot \text{BLEU}(y, y^*) + \omega_3 \cdot \Phi(y, S_T),$$

where $\omega_1, \omega_2, \omega_3 \geq 0$are weights, COMET provides a neural semantic-adequacy estimate, BLEU provides a reference-based lexical-overlap signal, and $\Phi(y, S_T)$is a style score produced by an external LLM-as-Judge protocol. The weights are treated as hyperparameters and tuned on a development set rather than fixed a priori.

The RL stage maximizes expected reward, $L_{\text{RL}}(\theta) = -\mathbb{E}_{y\sim P_\theta}[r(y)]$, and is combined with the supervised maximum-likelihood objective $L_{\text{MLE}}(\theta)$through a convex mixture:

$$L_{\text{total}}(\theta) = \lambda\, L_{\text{RL}}(\theta) + (1-\lambda)\, L_{\text{MLE}}(\theta), 0 \leq \lambda \leq 1,$$

where λ controls the trade-off between following human references and directly optimizing the style-sensitive reward.

Assumptions and limitations of the formulation

Three assumptions of this formulation are made explicit:

First, the three quality components (Fidelity, StyleMatch, Fluency) are assumed to be approximately separable. In practice they interact (a translation may sacrifice fluency to match an archaic register) but the additive form is retained for tractability.

Second, $\Phi(y, S_T)$ is assumed to provide a usable scalar estimate of stylistic match. Recent work on literary translation evaluation suggests that this assumption is not unconditional [15], and the project does not treat Φ as ground truth; it is used as a noisy signal alongside stylometric features and reference-based metrics.

Third, the formulation describes the optimization target. The two non-parametric strategies in this project (Adaptive Few-Shot Prompting and the unadapted reference model) do not update θ, but their outputs are evaluated against the same utility components, which makes cross-strategy comparison well-defined.

2.2 SIGNIFICANCES OF THE RESEARCH PROBLEM

The significance of this research has both methodological and applied dimensions. From a methodological perspective, the project examines how three distinct families of LLM adaptation(supervised fine-tuning via PEFT [20, 21], in-context learning via retrieval-based AFSP [18], and reinforcement learning with a style-sensitive reward [2, 3]) behave when applied to the same low-resource literary translation task under matched conditions. Each of these families has been studied individually in prior work, but the relative contribution of each to stylistic fidelity, and the trade-offs between them in terms of compute cost, data requirements, and stylistic control, are not well characterized in the published literature for this kind of corpus. A controlled side-by-side comparison on a single corpus contributes empirical evidence about these trade-offs that is not available from studies examining one method in isolation.

The research also intersects with several open questions in NLP that have been highlighted in recent shared-task findings and metric studies [10, 11], including low-resource adaptation in stylistically constrained domains, retrieval-based control of inference-time style, the design of reward signals for translation, and the application of style-sensitive evaluation methods to literary text. The project does not aim to resolve any of these questions in general terms. The comparative results it produces are intended to be informative for similar low-resource literary translation tasks in which a target stylistic register must be preserved, although the extent to which the findings generalize beyond the Bahá'í corpus is itself a question that the project can only partially address.

2.3 GAP IDENTIFICATION AND THE RESEARCH PROBLEM

The literature reviewed in Section 1.2 establishes that LLM-based machine translation has made substantial progress on general-domain translation, that neural systems can in principle respond to stylistic signals, and that LLMs can outperform dedicated MT systems in creative or literary domains under appropriate guidance. Despite this progress, three gaps are particularly relevant to the present project. They are summarized in Section 1.2 and stated in full here.

Gap 1 — Absence of a controlled comparison across adaptation strategies

LLM adaptation for translation can be grouped into three families: supervised fine-tuning, in-context learning with retrieved exemplars, and reinforcement learning with a style-sensitive reward. Each family has been studied individually (supervised fine-tuning via parameter-efficient methods [20, 21], in-context learning with retrieval-based exemplar selection [6, 8, 18], and reinforcement learning approaches for translation [2, 3]) and individual studies report results that support the usefulness of each method in particular settings. What the literature does not contain is a controlled side-by-side comparison of all three families applied to the same corpus, with the same evaluation protocol, in a low-resource literary setting. As a result, claims about the relative contribution of each family to stylistic fidelity, and the trade-offs between them in terms of compute cost, data requirements, and stylistic control, are made on the basis of cross-study comparisons that involve different corpora, language pairs, base models, and evaluation methods.

This gap matters because the three families operate through fundamentally different mechanisms (parameter update from references, inference-time conditioning, and parameter update from a reward signal) and a study that examines them under matched conditions can isolate the contribution of the adaptation signal in a way that cross-study comparisons cannot.

Gap 2 — Limited work on low-resource literary translation involving Persian and Arabic

Most published work on LLM-based translation concentrates on high-resource language pairs (typically involving English, French, German, or Chinese) or on general domains (news, web text, conversational data). Persian and Arabic appear in some multilingual benchmarks but are not the focus of style-aware translation work, and the specific case of literary or scriptural material with Persian and Arabic source segments occurring within the same document is rarely treated [9]. The two source languages share the Perso-Arabic script and have substantial lexical overlap through historical borrowing, but their grammatical systems are typologically distinct. Authoritative translators of Bahá'í material, including Shoghi Effendi, produced English texts in which a single stylistic register is maintained across mixed-language source passages. Existing LLM adaptation methods have not been evaluated against this combined challenge.

This gap matters because adaptation methods that work for high-resource pairs may not transfer to low-resource literary material, and methods that work for monolingual source material may not transfer to mixed-language source material. Documenting how the three adaptation families perform on this specific combination is a precondition for understanding where they generalize and where they do not.

Gap 3 — Behaviour of style-sensitive evaluation methods on low-resource literary translation

Reference-based metrics such as BLEU and TER are known to be insufficient on their own for assessing tone, register, and literary appropriateness [11]. More recent style-sensitive evaluation approaches (including LLM-as-Judge protocols [12, 15] and objective stylometric measures such as lexical density and type–token ratio) have been proposed, but their behaviour on low-resource literary translation is not yet well characterized [14, 15]. In particular, the agreement between automatic semantic metrics, stylometric features, and LLM-based judgments in this setting is largely an empirical question, and is itself one of the outcomes the present project is designed to report.

Statement of the research problem

The three gaps motivate the research problem stated at the beginning of Section 2: how can open-source LLMs be adapted, using a combination of Parameter-Efficient Fine-Tuning, Adaptive Few-Shot Prompting, and a bounded Reinforcement Learning stage with a style-sensitive reward, to produce English translations of low-resource, mixed Persian and Arabic literary texts that preserve both semantic content and a specified translator register? The project addresses this problem through a comparative experimental study under matched conditions, with the contribution framed as documentation of trade-offs rather than identification of a single optimal method.

2.4 Research questions/research objectives/hypothesis

The project is organized around four research questions.

RQ1. When the three LLM adaptation strategies (Parameter-Efficient Fine-Tuning (PEFT/LoRA), retrieval-based Adaptive Few-Shot Prompting (AFSP), and a bounded Reinforcement Learning stage with a style-sensitive reward (RLSF, implemented with PPO)) are applied to the same low-resource Persian/Arabic literary corpus under matched conditions, how do they compare in terms of semantic adequacy, stylistic fidelity to Shoghi Effendi's scriptural register, and compute and data cost?

RQ2. To what extent does retrieval-based exemplar selection in the AFSP stage, and reward-driven policy update in the RLSF stage, shift model outputs toward the target stylistic register relative to a PEFT-only baseline? In particular, can quantifiable shifts in stylometric features (lexical density, type–token ratio, register markers) be detected, and how do these shifts relate to LLM-as-Judge style scores?

RQ3. How does the composition of the reward function $r(y) = \omega_1 \cdot \mathrm{COMET} + \omega_2 \cdot \mathrm{BLEU} + \omega_3 \cdot \Phi(y, S_T)$affect the behaviour of the RLSF stage? Specifically, how sensitive are the trained policy's outputs to the choice of weights $\omega_1, \omega_2, \omega_3$, and which weighting produces the most favourable trade-off between semantic adequacy and stylistic fidelity?

RQ4. What is the relationship between the three evaluation components used in this study (neural semantic metrics, stylometric features, and LLM-as-Judge scoring) when applied to the same translation outputs? Where do they agree and where do they disagree, and what does this suggest about their joint use in evaluating low-resource literary translation?

Research Objectives

To address these questions, the project will:

- **Prepare and document the experimental corpus.** Curate and align the parallel corpus of Bahá'í scriptures in Persian and mixed Persian/Arabic with their authorized English translations by Shoghi Effendi; document the corpus size, alignment method, mixed-language handling, and train/development/test split.
- **Implement and adapt the PEFT/LoRA baseline.** Apply Parameter-Efficient Fine-Tuning with LoRA adapters to a selected open-source multilingual LLM on the prepared corpus, and produce a domain-adapted checkpoint that serves both as the standalone PEFT baseline and as the initial policy for the RLSF stage.

- **Implement and apply the AFSP pipeline.** Construct a retrieval index over the target-language portion of the corpus, develop the prompt template, and run inference-time generation with retrieved style-bearing exemplars under a fixed shot-count configuration with sensitivity analysis on the number of shots.

- **Implement and apply the RLSF stage under bounded conditions.** Initialize the policy from the PEFT-adapted checkpoint, configure the reward function $r(y)$combining COMET, BLEU, and an LLM-as-Judge style score Φ, and run PPO training under a declared compute budget with a fixed cap on the number of optimization steps and a KL-divergence constraint against the reference policy. A best-of-N reranking fallback, described in Section 8.3, is reserved for the case where PPO training fails to converge under the declared budget.

- **Evaluate all systems within a combined framework.** Apply the three evaluation components to outputs from the three adapted systems together with an unadapted base-model baseline, and report results with appropriate statistical tests for system-level comparisons.

- **Document the comparative findings.** Synthesize the results into a written account of how the three adaptation strategies trade off semantic adequacy, stylistic fidelity, and cost, including the points at which the methods fall short and the points at which the evaluation components agree or disagree.

Hypotheses

Rather than committing in advance to a single predicted ranking of the three methods, the project states the following testable hypotheses, each of which can be supported or refuted by the experimental results.

H1 (effect of adaptation on stylistic fidelity). Compared to the unadapted base-model baseline, at least one of the three adaptation strategies will produce a statistically significant improvement in LLM-as-Judge style scores on the held-out test set, evaluated at $\alpha = 0.05$under paired bootstrap resampling. *Rationale*: This is the minimal claim required for the project to have a positive result; failure to reject the null would itself be an informative finding about the difficulty of the task on this corpus.

H2 (relative behaviour of AFSP and RLSF). The AFSP stage and the RLSF stage will produce qualitatively different output profiles relative to the PEFT baseline: AFSP outputs will be more sensitive to retrieved exemplars at the level of lexical choice and will vary more across inputs, while RLSF outputs will show more uniform shifts across the test set toward the reward-favoured register. *Rationale:* The two methods operate through different mechanisms — inference-time conditioning on per-input exemplars versus parameter update from a globally optimized reward — and this difference is expected to be visible in the variability of stylometric features across outputs.

H3 (reward-weight sensitivity in RLSF). Varying the reward weights $\omega_1, \omega_2, \omega_3$over a small grid will produce measurable differences in the RLSF-trained policy's outputs, indicating that the reward composition is a substantive design choice rather than a free parameter. In particular, increasing $\omega_3$relative to $\omega_1$is expected to shift outputs toward higher LLM-as-Judge style scores at some cost to COMET. *Rationale:* The reward determines which translations PPO assigns higher return to, so changes in the weights are expected to change the optimization target visible in the outputs.

H4 (evaluation-component agreement). The three evaluation components (COMET, stylometric features, LLM-as-Judge) will exhibit substantial but not complete agreement when applied to the same outputs; in particular, COMET and stylometric features are expected to capture partially overlapping signals (semantic adequacy and register-relevant lexical patterns respectively), while LLM-as-Judge is expected to provide additional information not fully predicted by either. *Rationale:* This is the central methodological question motivating the use of three evaluation components rather than one.

The hypotheses are framed so that each can be supported or refuted by the experimental data without depending on the outcomes of the others. The project does not pre-commit to which adaptation method will perform best on the primary stylistic-fidelity metric.

3 Background and Preliminaries

This project works with parallel corpora, where Persian and Arabic source texts are aligned with high-quality English translations. These are not simple everyday sentences; they are literary and religious texts with strong stylistic expectations.

- Resources:

  Bahá'í scriptures originally revealed in Persian and Arabic with authorized English translations by Shoghi Effendi,

These texts are important because they embody distinctive writing and translation styles that the system aims to imitate.

- Style

  The style is the choice of words, sentence structure, tone, and overall "voice."

- Style can change depending on:

  - how formal or informal the situation is,
  - who the audience is,
  - and the personal habits and preferences of the writer or translator.

In this project, several dimensions of style are important:

1- Formality and Register
   - Bahá'í scriptures and similar texts generally require a formal, elevated register, often with precise, dignified, and sometimes archaic language.

- Informal language (e.g. contractions, slang) would feel out of place.
- The model must learn to keep the translation in the appropriate level of formality.

2- Individual / Personal Style
- Different translators have different "voices" and linguistic habits.
- For example, Shoghi Effendi's translations of Bahá'í texts are highly formal and scriptural in tone with different styles of Bahá'í texts
- The project aims to capture translator-specific styles, not just a generic "good English" style.

3- Genre and Poetic Language
- The data used in this project is therefore both linguistically rich and stylistically demanding.

To understand the technological part of the project, it is useful to briefly review how machine translation has developed.

- Early MT

  - The earliest systems were rule-based, relying on hand-written grammatical and lexical rules.
  - They worked reasonably well in limited domains but did not scale to open-domain, nuanced language.

- Statistical and Early NMT

  - Later, statistical machine translation (SMT) used large bilingual corpora and probabilistic models to learn which translations were likely.
  - SMT improved flexibility but still struggled with long-range context and fluency.
  - Then came Neural Machine Translation (NMT), which uses neural networks (especially encoder–decoder architectures) to model the translation process end to end.

- The Transformer

  - A major step forward was the Transformer architecture, which replaces recurrent networks with self-attention mechanisms.
  - Attention allows the model to focus on the most relevant parts of the input sentence when generating each output token.
  - Transformers are now the standard architecture behind modern NMT systems and LLMs.

- Context and Document-Level Translation

  - Traditional MT often translated sentences in isolation, ignoring surrounding context.
  - However, real texts – especially literary and religious ones – depend heavily on document-level context, including consistency of terminology, discourse structure, and narrative flow.
  - Recent research has therefore focused on using document-level or multi-sentence context to improve translation quality, something that is especially important for this project's data.

Large Language Models (LLMs) and Adaptation

Modern Large Language Models sit on top of the Transformer architecture and are central to the approach taken in this project. LLMs such as GPT-style models are trained on massive text corpora and can generate coherent, fluent text, translate between languages, and perform many other tasks through prompting.

They are promising for literary translation because they:

- can produce more natural, human-like sentences,
- can in principle adapt their style when given appropriate instructions or examples.

However, they also have limitations:

- their performance is often weaker on low-resource languages like Persian, especially for specialized domains,
- they are not automatically tuned to mimic specific translator styles,
- they sometimes produce confident but stylistically or semantically inappropriate output.
- Because of this, LLMs must be adapted carefully to the task.

- Adaptation Strategies

This project focuses on three main strategies for adapting LLMs to style-aware literary translation:

1- Fine-Tuning

   Fine-tuning means taking a pre-trained LLM or NMT model and training it further on the specific data used in this project.

   - The model is given pairs of inputs and outputs:

     - Input: Persian or Arabic passages from the Bahá'í writings
     - Output: their authorized English translations

   - During training, the model learns:

    - how meaning is transferred from Persian/Arabic to English in this domain,
    - what kind of vocabulary, phrasing, and sentence structure are typical of these translations,
    - how to maintain a consistently formal, scriptural tone.

A special style or translator signal (for example, a token indicating "authorized Bahá'í style") can be added to the input to guide the model toward the right voice.

Fine-tuning helps the model move from "generic translation" to something closer to the authorized style found in the corpus.

2- In-Context Few-Shot Learning and Prompting

In-context learning does not change the model's parameters. Instead, it changes what we put in the prompt.

- The idea is to show the model a few examples of the desired style inside the input, for example:

    - a short Persian or Arabic passage and its authorized English translation,
    - followed by a new passage that needs to be translated "in the same style."

- The model "sees" these examples and continues in a similar way for the new text.
- Style-focused prompting can include:
    - passages from authorized English Bahá'í translations as style examples,
    - explicit instructions, such as:
        - "Use a formal, reverent, and scriptural tone."
- This approach is flexible because:

    - it does not require retraining the model for each new style,
    - different styles can be activated by changing the examples in the prompt.

3- Reinforcement Learning (RL)

Reinforcement learning adds an extra layer, the model is not only trained to imitate examples, but also to maximize a reward score.

- The process works in three steps:
  1- The model generates a translation for a given passage.
  2- Another component (a reward function or a separate model) scores this translation. The score can reflect:
        - how well it preserves the meaning,
        - how fluent the English is,
        - how closely it matches the desired style.

3- The translation model is updated in a way that increases the chance of producing high-scoring translations in the future.

This method allows direct optimization for:

- semantic fidelity,
- stylistic fidelity,
- or a weighted combination of both.

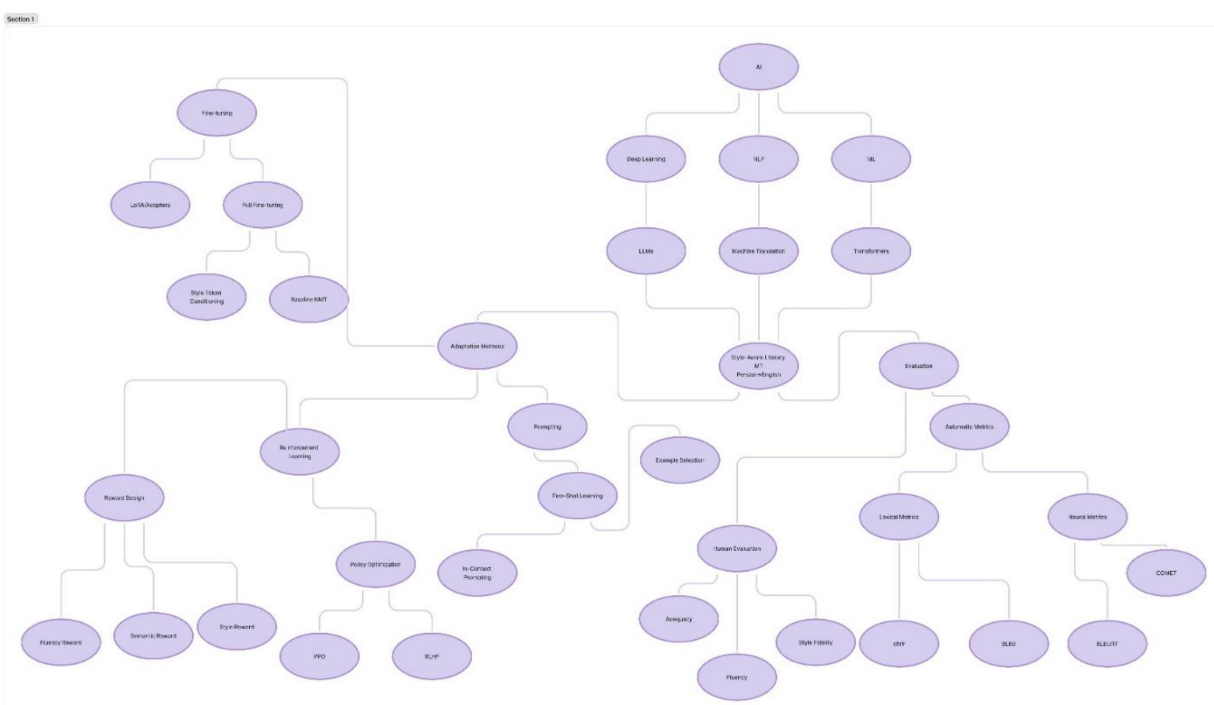
- *RL is especially useful*

we know what a “good” translation looks like (through metrics or human scores), but it is hard to express all requirements in a single simple loss function.

- *Evaluation*

To see whether these adaptation strategies actually improve translation quality, the project uses a combination of automatic metrics and human evaluation. Automatic metrics (such as BLEU, and COMET) give a rough, quantitative comparison between different systems. Human evaluation focuses on how well the translation preserves the meaning, how fluent and natural the English is, how closely the style matches the tone and register expected in authorized Bahá’í translations.

3.1 TREE OF THE KNOWLEDGE AND THE SPECIFIC PROBLEM



4 Preliminary Literature Review

4.1 High ranking journals, conferences, researchers

This subsection identifies the main academic venues and research lines that ground the present project. Detailed reviews of the specific contributions of each cited work are given in Section 4.2 (problem-side review) and Section 4.3 (methodology-side review).

Main publication venues

The project draws on work published in a small number of established venues in computational linguistics and machine translation. The *Annual Meeting of the Association for Computational Linguistics* (ACL) and the *Conference of the North American Chapter of the ACL* (NAACL), together with their Findings tracks, are the principal conference venues for work on translator-style modelling, LLM adaptation, evaluation methodology, and prompting for translation [3, 5, 15, 19]. The *Conference on Machine Translation* (WMT) is the dedicated venue for translation benchmarking and shared tasks. The WMT24 findings report [10] is used in the present project as the reference point for the current state of MT evaluation and the limitations of current systems on stylistically demanding material, and Zhang et al.'s WMT 2023 comparison of prompting, few-shot learning, and QLoRA fine-tuning [22] provides the closest published methodological analogue to the present project's three-way design.

For foundational machine-learning method work, the *International Conference on Learning Representations* (ICLR) and *Conference on Neural Information Processing Systems* (NeurIPS) publish the parameter-efficient fine-tuning papers used in the methodology, specifically Hu et al.'s LoRA [20] and Dettmers et al.'s QLoRA [21].

Beyond computational linguistics proper, work on style preservation in sacred and literary corpora has appeared in interdisciplinary journals bridging NLP and humanities-oriented text analysis. *Humanities and Social Sciences Communications* publishes studies including Gao et al. on Chinese classical poetry [7] and Gaanoun and Alsuhaibani on Quran translation [16]; the *Journal of King Saud University — Computer and Information Sciences* has published related work on emotion preservation in Quran translation [17].

Three research lines relevant to the project

The cited literature can be organized into three lines of research that together cover the problem space addressed by this project.

The first concerns MT evaluation and benchmarking in the LLM era. Work in this line examines what current MT systems can and cannot do, defines evaluation protocols, and reports the limitations of standard automatic metrics. The WMT24 findings report [10] is the central document in this line for the current period, presenting evidence that MT is not solved in stylistically demanding or low-resource settings even with LLM-based systems. Related work examines the behaviour of fine-tuned metrics in unseen domains [11] and the use of AI-assisted protocols in human MT evaluation [12], and recent studies of LLM-based literary translation evaluation report substantial disagreement between human and automatic judgments [14, 15]. The present project takes its framing of evaluation as a combination of neural semantic metrics, stylometric features, and judgment-based assessment from this line.

The second concerns LLM adaptation and reward-based optimization for translation. Work in this line studies how pre-trained LLMs can be adapted to translation tasks through parameter-efficient methods, prompting strategies, and reward-based training. Parameter-efficient adaptation methods used in the PEFT/LoRA stage of this project are documented in foundational ICLR and NeurIPS work [20, 21] and have been applied to translation in recent comparative studies [13, 22]. Work on self-refinement and structured reward composition for MT [2, 3] provides the reward-composition pattern that the present project adapts for the bounded RLSF stage.

The third concerns style modelling for literary and sacred-text translation. Work in this line examines the specific challenges of preserving register, tone, and emotional semantics in translation of stylistically dense source material. Studies on Quran translation [16, 17] and on Chinese classical poetry [7, 19] are the closest published analogues to the present project's setting, in that they share the central concern of preserving a target stylistic register that is constitutive of meaning rather than incidental to it. Foundational work on modelling translator style in NMT [5] and on retrieval- and prompting-based style control [6, 8, 18] supplies the modelling vocabulary used in the AFSP stage. The present project does not replicate any of these studies directly; rather, it draws on the methodological patterns established in this line and applies them to a previously unaddressed corpus and language combination.

Key authors

The cited work in these three lines includes several authors whose publications recur in the project's reading. Within the MT evaluation line, **Tom Kocmi** and **Vilém Zouhar** are recurring authors whose work [10, 11, 12] supplies the protocols and findings the project's evaluation framework builds on. Within the adaptation and reward-optimization line, **Zhaopeng Feng and co-authors** [2, 3] provide the reward-composition pattern adapted in this project, and **Edward Hu et al.** [20] and **Tim Dettmers et al.** [21] provide the parameter-efficient fine-tuning methods used in the PEFT stage. Within the literary and sacred-text line, the work of **Mohammed Alsuhaibani and co-authors** [16, 17] on Quran translation is the closest published analogue in domain, and **Shanshan Wang and co-authors** [6, 8, 19] supply the style-control and prompting framework used in the AFSP stage.

4.2 Literature review on the problem, gap identification, and the significance of the problem

This subsection reviews the empirical findings in the literature that support the three gaps identified in Section 2.3. The function of the review is to show what specific studies have reported on each problem area, rather than to restate the gap argument.

Evidence on stylistic limitations of current MT systems

Several recent studies report that current MT systems, including LLM-based systems, fail to preserve stylistic register in literary and stylistically dense translation tasks. Gao et al. [7], evaluating ChatGPT, Google Translate, and DeepL on Chinese classical poetry, report that all three systems struggle with the imagery, rhythm, and cultural density characteristic of literary source material, and that translation quality on such texts cannot be reliably captured by standard reference-based metrics. Wang, Wong, Yao, and Chao [19] reach a related conclusion in their ACL

2024 study of ChatGPT for poetry translation: improving stylistic quality requires deliberate prompt engineering and is not produced by default. Earlier work on translator-style modelling in NMT by Wang, Hoang, and Federico [5] established that translator-specific stylistic features can be modelled when training data is conditioned on translator identity, but this work predates the LLM era and was conducted on high-resource pairs. In the Quran translation literature, Gaanoun and Alsuhaibani [16] report that the seven reputable English translations of the Quran they study align with one another more than with the Arabic source on sentiment-preservation measures, and Almarzoqi and Alsuhaibani [17] extend this to emotion preservation and find that even reputable translations capture Quranic emotional semantics only at a moderate level. The shared finding across this literature is that stylistic and emotional preservation in translation of stylistically dense source material is a measurable problem on which current systems and translations both fall short.

Evidence on low-resource and mixed-language translation difficulty

The literature on low-resource translation with LLMs reports that the gains LLMs deliver on high-resource pairs do not transfer uniformly to the low-resource setting. Iyer et al. [9], in their AmericasNLP submission, document the difficulty of very low-resource translation even with strong base models and report that careful adaptation is required to obtain usable output. Hendy et al. [23] in their broader GPT-MT evaluation find that GPT-family models perform well on some high-resource directions but remain limited on low-resource and non-English-centric tasks. Zhang et al.'s WMT 2023 comparison [22] of prompting, few-shot learning, and QLoRA fine-tuning on French–English provides the closest published methodological analogue to the present project, and reports that QLoRA fine-tuning can substantially outperform prompting on that pair. Zebaze, Sagot, and Bawden [13] address low-resource translation through compositional methods. The present project's source domain — Persian and Arabic literary text with mixed-language passages — is not represented in this literature; the closest treatments are general low-resource studies and Arabic-only studies on Quran translation [16, 17], neither of which addresses the mixed-language case.

Evidence on the limitations of current evaluation methods

The literature on MT evaluation in the LLM era reports a consistent set of findings: reference-based automatic metrics are insufficient for stylistic and literary translation, but the proposed alternatives have their own limitations. The Kocmi et al. WMT24 findings report [10] explicitly frames the current state of the field as "the LLM era is here but MT is not solved yet," with errors visible across domains under professional human annotation. Zouhar et al. [11] report that fine-tuned MT metrics struggle on unseen domains, which limits the transferability of metric-based evaluation across corpora. Zouhar, Kocmi, and Sachan [12] develop AI-assisted human evaluation protocols and report that pre-filling error spans with automatic estimates can substantially reduce annotation time, indicating that the design of the evaluation process matters as much as the choice of metric. For literary translation specifically, Zhang, Zhao, and Eger [15] introduce LITEVAL-CORPUS and report that complex evaluation schemes, including MQM, can be inadequate for literary settings, and that human and automatic judgments disagree substantially in this domain; Zhang et al. [14] propose LITRANSPROQA as a question-answering-based evaluation alternative. The cumulative finding across this literature is that evaluation methodology for stylistic and literary translation is itself an open research area, and that no single metric is currently positioned as the consensus answer.

Cumulative significance

The findings reviewed above support the three gaps stated in Section 2.3 without restating them: empirical evidence exists that current MT systems struggle with stylistic preservation, that low-resource and mixed-language literary translation is a relatively unaddressed setting, and that the evaluation methods proposed as alternatives to BLEU are themselves still being characterized. The significance of the present project is that it addresses these three problem areas in a single comparative study on a corpus that has not, to our knowledge, been used in published LLM-based translation research.

4.3 Literature review on the solutions and the methodologies

This subsection reviews the methodological literature most directly relevant to the three adaptation strategies and the evaluation components used in this project. The function of the review is to show what methods have been proposed and what the published literature reports about them, not to argue that one method is uniquely correct.

Parameter-efficient fine-tuning for translation adaptation

Parameter-efficient fine-tuning (PEFT) has become a standard approach for adapting large language models on small, specialized corpora without the cost of full fine-tuning. The two methods most relevant to the present project are LoRA, introduced by Hu et al. [20], which inserts low-rank trainable update matrices into the attention layers of a frozen base model, and QLoRA, introduced by Dettmers et al. [21], which combines LoRA with 4-bit quantization of the base model to allow adaptation of large models on modest hardware. These methods have been applied to translation specifically: Zhang et al.'s WMT 2023 study [22] compares zero-shot prompting, few-shot prompting, and QLoRA fine-tuning on a single language pair and reports that QLoRA can substantially outperform prompting under matched conditions, providing the closest published methodological analogue to the present project's three-way comparison. Compositional approaches to low-resource translation with LLMs are described by Zebaze, Sagot, and Bawden [13]. The present project uses LoRA (with QLoRA quantization if compute constraints require it) as the parameter-efficient adaptation method for both the PEFT baseline and the initialization of the RLSF stage.

In-context learning and retrieval-based prompting for style control

A second line of methodological work studies how LLMs can be conditioned on stylistic targets at inference time, without parameter updates. Wang, Hoang, and Federico [5] established that translator-specific style can be modelled in NMT when training data is conditioned on translator identity; subsequent LLM-era work has shifted toward inference-time conditioning. Three distinct methods are particularly relevant here, and the present project treats them as related but not identical:

- **Wang et al.'s style-activation prompting** [6] proposes prompting strategies that activate stylistic patterns latent in pre-trained models without retrieval.
- **Wang et al.'s style-matching approach** [8] addresses the gap between zero-shot and few-shot translation by selecting exemplars that match the target style.
- **Tang et al.'s Adaptive Few-Shot Prompting (AFSP)** [18] proposes a retrieval-based exemplar-selection framework for translation with pre-trained language models.

The present project follows the AFSP pattern of [18] retrieval of stylistically relevant exemplars and dynamic insertion into the prompt and treats [6] and [8] as related precedents rather than as the same method. Wang, Wong, Yao, and Chao [19], in their ACL 2024 study of poetry translation with ChatGPT, report that explanation-assisted prompting can improve literary translation outputs and that prompt design matters substantially for stylistic quality. The shared finding across this line is that in-context conditioning can shift stylistic outputs but that exemplar selection and prompt design are themselves substantive design choices.

Reinforcement learning and reward composition for translation

A third methodological line studies the use of reward-based optimization to address translation quality dimensions that are not well captured by token-level supervised losses. Feng et al.'s TEaR system [3] proposes a translate–estimate–refine structure that uses model-internal feedback to improve translation outputs at test time. Feng et al.'s MT-R1-Zero work [2] proposes a rule-and-metric mixed reward composition for translation that combines reference-based and learned-metric signals, and the reward composition pattern used in the present project's RLSF stage is adapted from this work. The present project uses Proximal Policy Optimization (PPO) as the RL algorithm, initialized from the PEFT-adapted checkpoint and constrained by a KL-divergence term against the reference policy, as described in Section 6.

It should be noted that the literature on reward-based optimization for translation is not settled. The WMT24 findings report [10] indicates that MT is not solved even with LLM-based systems, and the relative merits of supervised fine-tuning, in-context learning, and reward-based optimization in low-resource literary settings are an open empirical question which is the question the present project is designed to address.

Evaluation methods used in the project

The evaluation components used in this project are all drawn from prior work. For semantic adequacy, COMET is a learned neural metric in widespread use in MT evaluation. For lexical overlap, BLEU is retained as a familiar reference-based baseline despite its known limitations [11]. For style-sensitive assessment, the project uses two components from the recent literature: objective stylometric features (such as lexical density and type–token ratio), which have been used in studies of literary and sacred-text translation [16, 17], and an LLM-as-Judge protocol, which has been developed and characterized in recent MT evaluation work [12] and applied to literary translation evaluation [14, 15]. The present project does not propose new metrics, new scoring criteria, or modifications to these components. Its contribution at the evaluation level is the joint application of these existing components to the same outputs and the analysis of their agreement.

5 Technology Review and Comparison

This section reviews the technical components required to implement the three adaptation strategies (PEFT, AFSP, RLSF/PPO) and the evaluation framework described in Section 6. The goal is to identify which models, libraries, and APIs are appropriate for each stage, and to make explicit the trade-offs and constraints introduced by each choice.

Base translation model: open-source vs. commercial LLMs

The base translation model is the LLM that is adapted in the PEFT and RLSF stages and prompted in the AFSP stage. Two categories of model are considered:

Commercial APIs (such as the GPT, Claude, and Gemini families) provide state-of-the-art zero- and few-shot performance and are often used as external baselines in MT evaluation [23]. Their two limitations for the present project are structural: the model weights are not accessible, which means parameter-efficient fine-tuning and reinforcement learning cannot be applied to them; and the per-token pricing of API calls makes them unsuitable for any process that involves large numbers of generations during training. Commercial APIs are therefore restricted in this project to two roles in which weight access is not required: as an unadapted reference point for translation quality, and as the LLM-as-Judge component of the evaluation framework and the reward function.

Open-source models (such as LLaMA, Mistral, and similar multilingual decoder-only models) provide full access to model weights and architecture, which is required for the PEFT and RLSF stages. The trade-off is that open-source models of comparable parameter count to top commercial systems typically lag in raw zero-shot translation quality, particularly for non-English-centric directions [23, 24]. For this project, full weight access is the binding requirement, and an open-source multilingual model in the 7B–8B parameter range will be selected as the base. The specific model choice will be finalized after a small-scale comparison of candidate models on the development set; the candidates considered are recent multilingual instruction-tuned models with documented Persian and Arabic coverage.

Parameter-efficient fine-tuning: LoRA and QLoRA

The PEFT stage uses LoRA [20] as the parameter-efficient adaptation method. LoRA inserts low-rank trainable matrices into the attention layers of a frozen base model, which reduces the number of trainable parameters by two to three orders of magnitude and makes adaptation feasible on consumer-grade GPUs. For a 7B-parameter base model, full fine-tuning is not feasible on the hardware available to this project; LoRA-based adaptation is.

If memory constraints during the RLSF stage exceed what LoRA alone can accommodate, QLoRA [21] will be used instead. QLoRA combines LoRA with 4-bit quantization of the base model and has been shown to enable fine-tuning of much larger models on consumer hardware with limited quality loss. The choice between LoRA and QLoRA is treated as a deployment-time decision rather than a methodological one; both are well-established techniques.

The reference implementation for both methods is the Hugging Face peft library, which is the standard open-source implementation and is the implementation that the cited papers [20, 21] and follow-up work [22] use.

Retrieval and prompting infrastructure for AFSP

The AFSP stage requires three components: a sentence-embedding model for indexing target-language exemplars, a vector index for nearest-neighbour retrieval, and a prompt-assembly pipeline that inserts retrieved exemplars into the model input.

For embeddings, a multilingual sentence-transformer (such as a paraphrase-multilingual-mpnet variant or a comparable model) provides adequate retrieval quality for stylistic similarity in English exemplars; the embedding model is used at index-construction time and at inference time and does not need to be fine-tuned.

For the vector index, FAISS is the standard open-source choice and is sufficient for the corpus size considered in this project; an in-memory index over a few thousand exemplars does not require more complex infrastructure.

For prompt assembly, the project uses direct string templating rather than a higher-level orchestration framework. A heavier orchestration framework (such as LangChain) was considered but is not required for the retrieval-and-prompt pipeline as described, and adopting it would introduce dependencies that exceed what the AFSP stage requires.

Reinforcement learning infrastructure for RLSF

The RLSF stage uses Proximal Policy Optimization (PPO) as the reinforcement learning algorithm. The reference implementation is the Hugging Face trl (Transformer Reinforcement Learning) library, which provides a PPO trainer compatible with peft-adapted models and supports the KL-divergence regularization against a reference policy that the RLSF stage relies on.

Three components are required to operate the RLSF stage:

- **The policy model**, initialized from the PEFT-adapted checkpoint produced in the previous stage.
- **The reference policy**, frozen at initialization and used to compute the KL-divergence penalty.
- **The reward function**, which combines COMET (computed with the unbabel-comet library), BLEU (computed with sacrebleu), and an LLM-as-Judge style score $\Phi(y, S_T)$.

The LLM-as-Judge component of the reward calls a commercial API (GPT-family or comparable). This is the most cost-sensitive technical decision in the project: each PPO training step requires scoring a batch of generations, and the API cost scales linearly with the number of generations scored. To control this cost, the RLSF stage operates under a declared compute budget specified in Section 8.3, which includes a maximum number of PPO steps, a maximum batch size, and a cap on total API expenditure. The use of an external API inside a training loop is acknowledged as a non-trivial operational constraint; it is the reason the RLSF stage is described throughout this proposal as "bounded."

Evaluation infrastructure

The evaluation framework uses three components, each with an established open-source or API-based implementation:

- **COMET** is computed using the unbabel-comet library with the wmt22-comet-da checkpoint, which is the current standard reference-based neural metric for MT.
- **Stylometric features** (lexical density, type–token ratio, and a small set of register markers) are computed with standard Python NLP tools (spaCy, nltk). No specialized stylometric library is required at the scale of the test set used here.

- **LLM-as-Judge scoring** uses the same commercial API as the RLSF reward, with a separate prompt template designed for evaluation rather than ranking. To avoid circularity between training and evaluation, the evaluation calls use a fixed prompt template and a held-out test set that is not used during RLSF training.

Summary of technology choices

The technology choices made in this project are summarized below. None of the components is novel; the design choice is which combinations are used at which stages and under what cost constraints.

Stage	Primary technology	Role	Notes on choice
Base model	Open-source multilingual LLM (7B–8B)	Translation model, adapted across all three stages	Full weight access required for PEFT and RLSF
PEFT stage	Hugging Face peft (LoRA, optionally QLoRA)	Domain adaptation	Standard reference implementation
AFSP retrieval	Multilingual sentence-transformer + FAISS	Exemplar selection	Lightweight in-memory pipeline
AFSP prompting	Direct string templating	Prompt assembly	Higher-level orchestration not required at this scale
RLSF training	Hugging Face trl (PPO)	Policy optimization	Compatible with peft-adapted policies; supports KL regularization
Reward — semantic	unbabel-comet (COMET)	Adequacy signal in reward	Standard MT metric
Reward — lexical	sacrebleu (BLEU)	Reference-overlap signal in reward	Familiar baseline despite known limitations
Reward — style	Commercial API (LLM-as-Judge)	Style signal in reward	Subject to per-call cost; bounded by Section 8.3 budget
Evaluation — semantic	unbabel-comet (COMET)	Held-out test scoring	Same library as reward, separate use
Evaluation — style	Stylometry (spaCy, nltk) + LLM-as-Judge	Held-out test scoring	Stylometry and LLM-as-Judge applied jointly

6 Proposed Methodology

The methodology of this project is a comparative experimental study of three Large Language Model adaptation strategies applied to the same low-resource literary translation task. The three strategies (Parameter-Efficient Fine-Tuning (PEFT/LoRA), Adaptive Few-Shot Prompting (AFSP), and a bounded Reinforcement Learning stage with a style-sensitive reward (RLSF/PPO)))are applied to the same open-source base model, trained or prompted with the same parallel corpus of Bahá'í scriptures, and evaluated within the same combined framework of neural semantic metrics, objective stylometric features, and LLM-as-Judge scoring. The unadapted base model is included as a reference point. The methodology is designed to be feasible under the resource constraints described in Section 5 and the compute and API budgets specified in Section 8.3.

The subsections of this chapter are organized as follows: Section 6.1 describes the high-level structure of the comparison and the role of each stage; Section 6.2 identifies the theoretical grounding for each method without proposing new theory; Section 6.3 specifies the model architecture and the formal optimization objective; Section 6.4 describes the algorithms used at each stage; Section 6.5 describes the dataset; and Section 6.6 describes the experimental design including independent and dependent variables, statistical analysis, and the held-out test set.

6.1 Proposed approaches

The project compares three adaptation strategies that operate through fundamentally different mechanisms. Each strategy is included because it represents a distinct family of LLM adaptation, and the comparison is designed to isolate the contribution of each mechanism under conditions that are otherwise matched.

PEFT/LoRA: parameter update from references. The first strategy is supervised fine-tuning of the base model on the parallel corpus using Low-Rank Adaptation [20]. LoRA updates a small number of trainable parameters inserted into the attention layers of the base model while keeping the original weights frozen. The expected effect is that the base model becomes anchored in the vocabulary, terminology, and surface patterns of the target domain. This stage produces a domain-adapted checkpoint that is used both as a standalone baseline and as the initialization point for the RLSF stage [22].

AFSP: inference-time conditioning. The second strategy applies no parameter updates. The base model is prompted at inference time with a small number of retrieved stylistically relevant exemplars drawn from the target-language portion of the corpus. The retrieval is performed by nearest-neighbour search over sentence embeddings, and the retrieved exemplars are inserted into the prompt according to a fixed template. This approach follows the pattern of Adaptive Few-Shot Prompting [18] and is related to earlier work on style-activation prompting and style-matching exemplar selection [6, 8]. The expected effect is that the base model's output is conditioned, per input, on the stylistic register of the retrieved exemplars.

RLSF: parameter update from a style-sensitive reward. The third strategy initializes the policy from the PEFT-adapted checkpoint and updates it using Proximal Policy Optimization (PPO) against a reward function that combines semantic, lexical, and style-sensitive signals. The reward function is described formally in Section 6.3 and follows the reward-composition pattern proposed by Feng et al. [2, 3]. The expected effect is that the policy is shifted, across the training distribution, toward outputs that the reward function favors, including outputs whose stylistic properties are not directly captured by token-level supervised losses. The RLSF stage is bounded in scale, with a fixed cap on training steps and total reward-evaluation calls, as specified in Sections 5 and 8.3.

Reference point. The unadapted base model, prompted with a minimal instruction and no exemplars, is included as a reference point. Its role is to provide a lower bound on translation quality without adaptation and to enable measurement of the absolute improvement contributed by each of the three adaptation strategies.

The four conditions (PEFT, AFSP, RLSF, and reference) are compared on the same held-out test set using the same evaluation framework, described in Section 6.6.

6.2 Proposed theories

The project is grounded in three established theoretical frameworks.

Neural Machine Translation as conditional sequence modelling. All three strategies treat translation as a conditional probability task $P_\theta(y \mid x)$over token sequences, with the model implemented as a decoder-only Transformer pre-trained on multilingual text. The present project extends this conditional formulation to style-aware translation as $P_\theta(y \mid x, S_T)$, where $S_T$is a fixed target stylistic register. This extension is realized differently in each stage: in PEFT and RLSF the conditioning is implicit in the parameters θ, and in AFSP it is realized through retrieved exemplars in the input.

Reward-based optimization for non-differentiable objectives. The RLSF stage relies on the framework of reinforcement learning from a scalar reward, in which a policy is optimized against an objective that need not be differentiable with respect to the model's outputs. This framework is the standard theoretical basis for using reward signals (including signals derived from human or model judgments) to influence the behavior of language models. In the present project, the reward function combines automatic metrics (COMET, BLEU) with a model-judged style score, and the theoretical role of the framework is to make the optimization of a style-sensitive composite objective well-defined.

Stylistics in translation studies. The project takes from translation studies and stylistics the position that in literary and scriptural texts, stylistic features such as register, cadence, and lexical choice are constitutive of meaning rather than incidental to it. This position is reflected in the project's choice to evaluate translations along a stylistic axis rather than along semantic adequacy alone, and in the decision to design a reward function whose style component is treated as substantive rather than ornamental. The project does not develop a theory of literary style; it operationalizes existing stylistic measures (lexical density, type–token ratio, judgment-based register scoring) and treats their joint behavior as the empirical object of study.

6.3 Proposed models/model formulation

This subsection specifies the model components used in the project and the optimization objective. The formal mathematical definition of the problem and the reward function is given in Section 2.1; this subsection refers to that formulation rather than restating it, and focuses on the architectural choices that operationalize it.

Translation model

The translation model used across all stages is an open-source multilingual decoder-only Transformer, selected from the candidate range described in Section 5. The same base model is used for all four experimental conditions (PEFT, AFSP, RLSF, and the unadapted reference), so that differences in output between conditions can be attributed to the adaptation mechanism rather than to differences in the underlying model.

Two architectural details are common across the adapted conditions:

- **LoRA placement.** Trainable low-rank update matrices are inserted into the query and value projection layers of each attention block, with the LoRA rank treated as a hyperparameter. The remaining base-model parameters are frozen. This configuration follows the standard LoRA setup [20] and is supported by the Hugging Face peft library.
- **Reference policy.** For the RLSF stage, a second copy of the PEFT-adapted model is held frozen and used to compute the KL-divergence penalty against the active policy. This is the standard configuration for PPO-based fine-tuning of language models and is the configuration implemented by the Hugging Face trl library.

The same tokenizer ships with the base model and is used unchanged across all stages.

Reward function (RLSF stage)

The reward function used in the RLSF stage is defined formally in Section 2.1. In summary, it is a non-negative weighted sum of three components:

$$r(y) = \omega_1 \cdot \mathrm{COMET}(x, y) + \omega_2 \cdot \mathrm{BLEU}(y, y^*) + \omega_3 \cdot \Phi(y, S_T),$$

where COMET provides a neural semantic-adequacy estimate, BLEU provides a reference-based lexical-overlap signal, and $\Phi(y, S_T)$is a style score produced by an external LLM-as-Judge component. The weights $\omega_1, \omega_2, \omega_3 \geq 0$are treated as hyperparameters, with sensitivity analysis on a small grid as specified in Research Question 3 (Section 2.4) and Hypothesis H3.

The reward-composition pattern — combining a semantic, a lexical, and a learned style signal — follows Feng et al. [2, 3]. The specific operationalization in this project (COMET + BLEU + judge-derived Φ) is project-specific but does not constitute a methodological contribution; the components are existing metrics applied in a familiar additive form.

LLM-as-Judge component for Φ

The style score $\Phi(y, S_T)$is produced by a budget-tier commercial LLM (as specified in Section 5) acting as an automated evaluator. It is not the translation model and does not share parameters with it. The judge receives the source segment x, the candidate translation y, and a fixed prompt template that defines the target stylistic register $S_T$in terms of features the judge is asked to score. The output is a scalar that is then incorporated into $r(y)$.

The use of an LLM-as-Judge in both the training reward and the final evaluation introduces a potential circularity. Three measures are used to limit this:

- The judge prompt template used during RLSF training is shorter and oriented toward ranking among candidates; the judge prompt template used at final evaluation is separate and oriented toward absolute scoring. Both templates are fixed at the start of their respective phases and are not tuned against the test set.
- The held-out test set used for final evaluation is not seen during RLSF training.
- Where feasible within the project's API budget, a confirmation pass on the final test set is performed with a judge model from a different commercial family, and the agreement between the two judges is reported. This is reported as a check on the primary judge's stability and is not required for the project's main results.

Recent work on LLM-based evaluation of literary translation [14, 15] indicates that judgments in this domain are not fully calibrated and that human–automatic disagreement is substantial. The judge component is therefore treated as a noisy stylistic signal that is useful in combination with other components, not as a ground-truth measure.

Optimization objective

The supervised training objective for the PEFT stage is the standard token-level maximum-likelihood loss $L_{\mathrm{MLE}}(\theta)$. The RLSF stage maximizes expected reward $\mathbb{E}_{y\sim P_\theta}[r(y)]$under a KL-divergence constraint against the reference policy, as implemented in the `trl` PPO trainer.

The two objectives are not blended into a single composite loss in this project; PEFT and RLSF are run as sequential stages rather than as a joint optimization. The convex-mixture formulation $L_{\mathrm{total}}(\theta) = \lambda L_{\mathrm{RL}}(\theta) + (1-\lambda)L_{\mathrm{MLE}}(\theta)$shown in Section 2.1 describes the relationship between the two objectives at the formulation level, but in practice $\lambda$is treated as a stage selector rather than as a tunable mixing coefficient: $\lambda = 0$for the PEFT stage and $\lambda = 1$for the RLSF stage, with the RLSF stage relying on the KL-divergence term to maintain proximity to the PEFT-initialized reference policy.

The AFSP stage does not update model parameters and therefore does not have an associated training objective. Its outputs are obtained by sampling from $P_\theta(y \mid \mathrm{prompt}(x, \text{retrieved exemplars}))$where the model is the unadapted base model.

6.4 Datasets

Real dataset

The project uses a parallel corpus of Bahá'í scriptures with English translations by Shoghi Effendi. The source texts are in Persian and contain interleaved passages in Arabic; the target texts are Shoghi Effendi's authorized English translations.

Alignment. Source–target alignment is performed at the sentence and paragraph level.

Mixed-language handling. Source segments containing both Persian and Arabic are retained as single segments; the model is responsible for handling the within-segment language mixing rather than treating Arabic spans as a separate translation task. This decision matches the way the authoritative English translations were produced, in which a single stylistic register is maintained across mixed-language passages.

Splits. The corpus is divided into training, development, and test partitions with approximate proportions **80% / 10% / 10%**. Splitting is performed at the document or section level rather than at the segment level, to reduce leakage of stylistic patterns from training documents into the test set. The split is fixed at the start of the experimental procedure and is not modified afterward.

Preprocessing. Standard text normalization is applied: Unicode normalization, removal of editorial metadata not present in the source–target alignment, and consistent handling of diacritics. No tokenization is performed at the corpus level; tokenization is delegated to the base model's tokenizer at training and inference time.

Synthetic dataset

No synthetic parallel data is generated for the training partition. The PEFT and RLSF stages are trained only on the authentic parallel corpus.

This decision reflects two considerations. First, the corpus is centered on a specific translator's authorized stylistic register, and synthetic translation pairs produced by an unadapted LLM would not reliably reproduce that register; using such pairs in training would risk diluting the very signal the project is trying to learn. Second, the bounded scope of the project does not include a separate validation effort to characterize the quality of synthetic data, which would be required before such data could be used responsibly in training.

6.5 Proposed Experiment Design

The experimental design implements the comparative study described in Section 6.1 under the resource constraints described in Section 5. This subsection specifies the experimental procedure, the variables and outcomes measured, the splits used, the evaluation procedure, the statistical analysis plan, and the criteria under which each hypothesis stated in Section 2.4 is considered supported or refuted.

Experimental procedure

The experiment proceeds in six stages, in order:

1. **Corpus preparation.** The parallel corpus described in Section 6.5 is segmented, aligned, and split into training, development, and test partitions. The mixed-language handling rules and the alignment method are documented; the splits are fixed at this stage and not modified afterward.
2. **PEFT/LoRA training.** The base model is fine-tuned with LoRA on the training partition. Hyperparameters (LoRA rank, learning rate, number of training steps) are tuned on the development partition. The output is a single PEFT-adapted checkpoint that serves both as the PEFT condition and as the initialization for the RLSF stage.

3. **AFSP retrieval index construction and prompt-template finalization.** A FAISS index is built over the target-language portion of the training partition using a multilingual sentence-transformer. The prompt template and the number of retrieved exemplars per input (k) are finalized on the development partition. The shot-count sensitivity analysis specified in Research Question 2 is conducted on the development partition only.

4. **RLSF training.** The policy is initialized from the PEFT-adapted checkpoint and trained with PPO under the bounded budget specified in Section 5 and Section 8.3. The reward weights ($\omega_1, \omega_2, \omega_3$)are selected on the development partition using a small grid as specified in Research Question 3.

5. **Test-set inference.** All four conditions (PEFT, AFSP, RLSF, and the unadapted reference base model) produce translations of the test partition under fixed decoding settings.

6. **Evaluation.** The three evaluation components (COMET, stylometric features, LLM-as-Judge) are applied to the four conditions' outputs on the test partition, as described below.

Each stage produces artifacts that are versioned and retained: the PEFT checkpoint, the AFSP index and prompt template, the RLSF checkpoint, and the test-set output files for all four conditions. These artifacts allow the evaluation stage to be re-run if any of the evaluation components are updated.

Variables and outcomes

Independent variables (controlled by the experimental design):

- *Adaptation strategy*: a four-level factor with levels {Reference, PEFT, AFSP, RLSF}.

- *Reward weights ($\omega_1, \omega_2, \omega_3$)*: varied over a small grid for the RLSF stage only, as specified in RQ3.

- *Number of exemplars k*: varied for the AFSP stage only, with sensitivity analysis on the development set.

Dependent variables (measured outcomes):

- *Semantic adequacy*: COMET score against the reference translation.

- *Lexical overlap*: BLEU score against the reference translation.

- *Stylistic fidelity*: LLM-as-Judge style score $\Phi(y, S_T)$.

- *Stylometric profile*: a vector of stylometric features (lexical density, type–token ratio, and a small set of register markers) computed on the system output.

- *Cost*: trainable parameter count for PEFT and RLSF; inference latency per input segment for AFSP; total reward-evaluation calls and API expenditure for RLSF.

The relationships measured are: the effect of adaptation strategy on each dependent variable (RQ1, RQ2); the effect of reward weights on the RLSF dependent variables (RQ3); and the pairwise agreement between the three evaluation components when applied to the same outputs (RQ4).

Evaluation procedure

Evaluation is conducted on the held-out test partition only. The development partition is used for hyperparameter selection (PEFT learning rate and rank, AFSP k, RLSF reward weights and PPO step count) and is not used in the final reported results. The training partition is used for parameter updates only and is not used for evaluation.

For each test segment, each of the four conditions produces one translation under fixed decoding settings (sampling temperature, top-p, and maximum length are held constant across conditions). The same reference translation by Shoghi Effendi is used for all reference-based metrics.

The three evaluation components are applied as follows:

- **COMET** is computed per segment and aggregated as the mean over the test set, with a paired bootstrap confidence interval at 95%.
- **BLEU** is computed at the corpus level using the sacrebleu configuration.
- **Stylometric features** are computed per segment and aggregated per condition; condition-level distributions are compared.
- **LLM-as-Judge style scoring** is computed per segment using the evaluation prompt template specified in Section 6.3; scores are aggregated as the mean per condition with a paired bootstrap confidence interval at 95%.

Statistical analysis plan

System-level comparisons between conditions are conducted using paired bootstrap resampling at the segment level with $\alpha = 0.05$, following standard practice in MT evaluation. The principal comparison is each adaptation condition (PEFT, AFSP, RLSF) against the unadapted reference; the secondary comparisons are pairwise between the three adaptation conditions.

For the evaluation-component agreement question (RQ4 and H4), the per-segment scores from COMET, the stylometric features, and the LLM-as-Judge are correlated pairwise using Spearman rank correlation, reported with 95% confidence intervals obtained by bootstrap. No claim of causal relationship between the evaluation components is made; the analysis is descriptive.

The hypotheses stated in Section 2.4 are considered supported as follows:

- **H1** is supported if at least one adaptation condition produces a statistically significant improvement in LLM-as-Judge style score over the reference at $\alpha = 0.05$under paired bootstrap.

- **H2** is supported if the variance of stylometric features across the test set differs between the AFSP and RLSF conditions in the direction predicted (AFSP higher variance, RLSF lower variance) at $\alpha = 0.05$.

- **H3** is supported if at least one pair of reward-weight configurations produces a statistically significant difference in the RLSF outputs on either the COMET or LLM-as-Judge dimension at $\alpha = 0.05$.

- **H4** is supported if the three evaluation components produce Spearman rank correlations in the substantial-but-incomplete range (interpreted as 0.4–0.8 inclusive); correlations outside this range — either weaker than 0.4 or stronger than 0.8 — would each be reported as a different finding about the joint use of these components.

Failure to support any individual hypothesis is reported as an informative result rather than as a project failure; the descriptive comparison of the four conditions on the test set is the primary contribution regardless of which hypotheses are supported.

7 Expected Results

This comparative experimental study is expected to provide evidence about which LLM adaptation strategy offers the most effective balance between semantic accuracy and stylistic fidelity for low-resource literary translation. Rather than assuming that one method will be universally superior, the study will compare Parameter-Efficient Fine-Tuning (PEFT/LoRA), Adaptive Few-Shot Prompting (ICL/StyleAP), and Reinforcement Learning with a Style-Sensitive Reward Function (RLSF/PPO) across multiple evaluation criteria.

Core Expected Findings

The main expected outcome is that each adaptation strategy will show different strengths and limitations. The PEFT/LoRA model is expected to provide a strong domain-specific baseline by improving the model's familiarity with the vocabulary, sentence structures, and translation patterns of the selected corpus. The ICL/StyleAP approach is expected to demonstrate whether carefully selected in-context examples can guide the model toward the desired stylistic register without additional training. The RLSF/PPO approach, if computationally feasible, is expected to provide the strongest stylistic control because it directly optimizes the model using a reward function that includes both semantic and stylistic criteria [2, 3, 11].

It is expected that the RLSF model may achieve higher scores on style-focused evaluation measures, especially the LLM-as-Judge score and stylometric indicators, because the reward function explicitly includes stylistic fidelity. However, this improvement may involve trade-offs in computational cost, training complexity, and possibly semantic accuracy. Therefore, the results will not only identify the best-performing model but also clarify the practical costs and limitations of each approach.

The ICL/StyleAP model is expected to perform competitively in cases where the retrieved examples are semantically relevant and stylistically close to the target register. This method may be especially useful in low-resource settings because it does not require model fine-tuning and can adapt dynamically at inference time. However, its performance is also expected to depend strongly on the quality and relevance of the selected examples [3, 15].

A final expected result of the study is a methodological roadmap for style-aware literary translation in low-resource domains. This roadmap will summarize which adaptation strategy is most suitable under different constraints, such as limited data, limited computing resources, need for stylistic precision, and acceptable inference cost.

Expected Theoretical and Methodological Contributions

The study is expected to provide practical evidence for the usefulness of reinforcement learning and reward-based optimization in style-aware machine translation. If the RLSF model improves stylistic fidelity without a major loss in semantic accuracy, the results will support the idea that complex stylistic goals can be partially represented through a mixed reward function combining semantic metrics, stylometric measures, and LLM-based evaluation [2, 3].

The study is also expected to contribute to the computational modeling of literary style. By comparing model outputs using stylometric indicators such as lexical density, type-token ratio, formality markers, and stylistic distance, the research may help identify which measurable linguistic features are most closely associated with the target dignified and scriptural style.

However, the results will be interpreted cautiously. The study will not claim to fully capture literary or sacred style, since such qualities are complex and partly subjective. Instead, it will evaluate whether existing LLM adaptation methods can improve measurable aspects of stylistic fidelity in comparison with generic LLM and commercial MT baselines.

Evaluation and Presentation of Results

The models will be evaluated using a multi-metric framework that includes semantic, stylistic, and efficiency-based measures. Semantic accuracy will be measured using metrics such as COMET and BLEU. Stylistic fidelity will be assessed through LLM-as-Judge evaluation and objective stylometric analysis. Efficiency will be considered through factors such as training cost, number of trainable parameters, inference time, and implementation complexity [17, 18, 20, 22].

The expected results will be presented using comparative tables and visualizations. A central comparison table will summarize the performance of the main systems: the unadapted LLM, the commercial MT baseline, the PEFT/LoRA model, the ICL/StyleAP model, and the RLSF/PPO model. The table will compare these systems across semantic accuracy, stylistic fidelity, stylometric distance, lexical density, and efficiency.

Additional visualizations may include a correlation plot comparing LLM-as-Judge scores with objective stylometric measures, and a chart showing how different reward function weights affect the trade-off between semantic accuracy and stylistic fidelity in the RLSF model. These visualizations will help determine whether the proposed evaluation framework provides consistent and interpretable results.

Interpretation and Discussion Focus

The discussion will focus on the comparative strengths and weaknesses of each adaptation strategy. If RLSF achieves the strongest stylistic performance, the analysis will examine whether this improvement justifies its higher computational cost and technical complexity. If ICL/StyleAP performs close to the fine-tuned models, the study may suggest that retrieval-based prompting is a practical alternative for low-resource literary translation. If PEFT/LoRA provides the best balance between quality and efficiency, it may be recommended as the most realistic baseline for similar projects.

The discussion will also examine the relationship between automatic metrics and stylistic evaluation. Since traditional metrics such as BLEU may not fully reflect literary quality, the study will analyze whether LLM-as-Judge scores and stylometric indicators provide a more informative assessment of style-aware translation quality. Finally, the linguistic analysis will identify which features, such as formality, lexical richness, sentence structure, or register consistency, most clearly distinguish the outputs of the different models.

Overall, the expected contribution of this research is not only a comparison of LLM adaptation methods, but also a clearer understanding of how style-aware translation systems can be evaluated and developed under low-resource conditions.

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